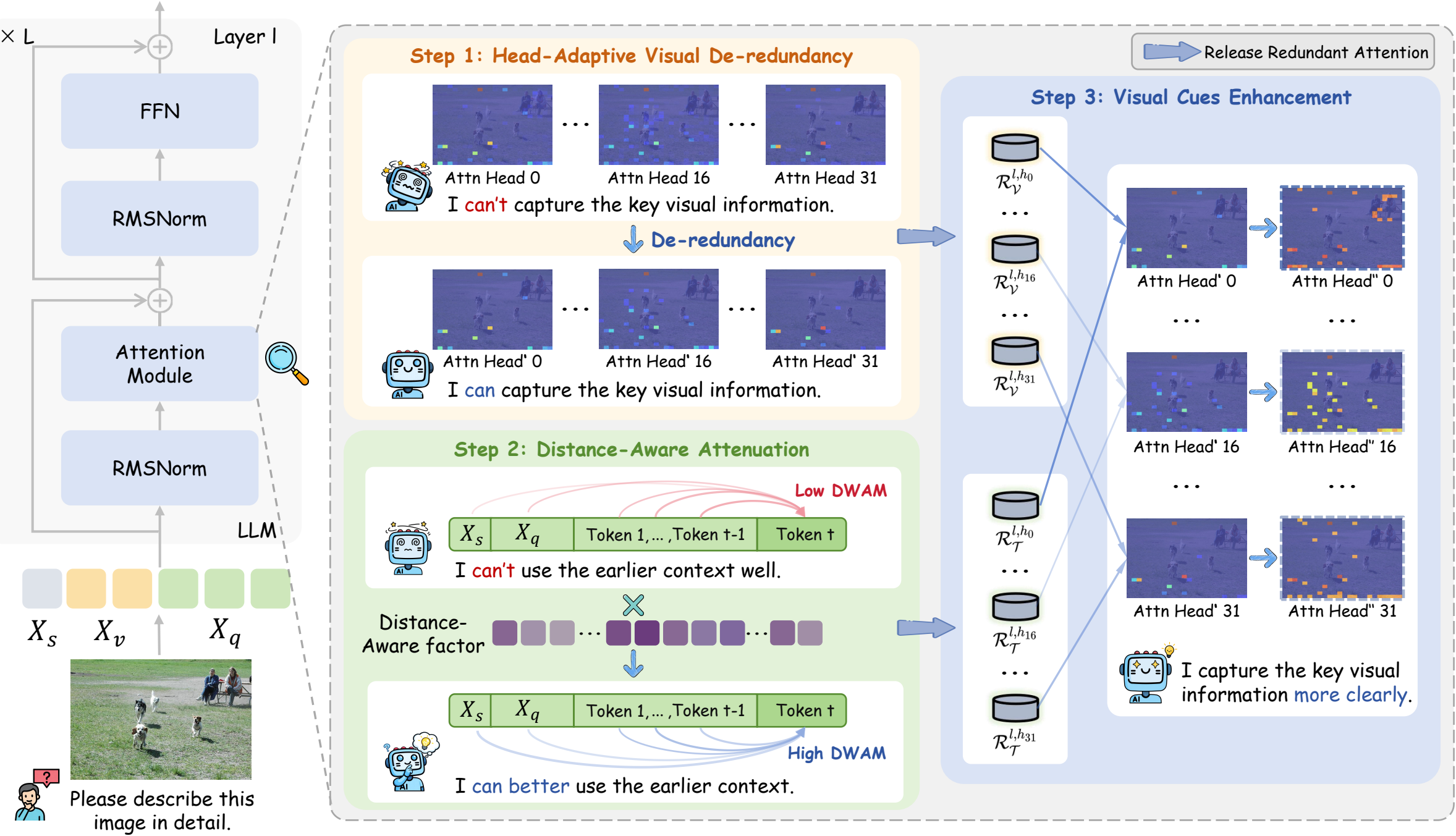

× L
Layer l
FFN
RMSNorm
Attention Module
RMSNorm
LLM
X_s X_v X_q
Please describe this image in detail.
Release Redundant Attention
Step 1: Head-Adaptive Visual De-redundancy
Attn Head 0
Attn Head 16
Attn Head 31
I can't capture the key visual information.
De-redundancy
Attn Head' 0
Attn Head' 16
Attn Head' 31
I can capture the key visual information.
Step 2: Distance-Aware Attenuation
Low DWAM
X_s X_q Token 1,... ,Token t-1 Token t
I can't use the earlier context well.
Distance-Aware factor
X_s X_q Token 1,... ,Token t-1 Token t
High DWAM
I can better use the earlier context.
Step 3: Visual Cues Enhancement
$\mathcal{R}_{\mathcal{V}}^{l,h_0}$
$\mathcal{R}_{\mathcal{V}}^{l,h_{16}}$
$\mathcal{R}_{\mathcal{V}}^{l,h_{31}}$
$\mathcal{R}_{\mathcal{T}}^{l,h_0}$
$\mathcal{R}_{\mathcal{T}}^{l,h_{16}}$
$\mathcal{R}_{\mathcal{T}}^{l,h_{31}}$
Attn Head' 0
Attn Head'' 0
Attn Head' 16
Attn Head'' 16
Attn Head' 31
Attn Head'' 31
I capture the key visual information more clearly.